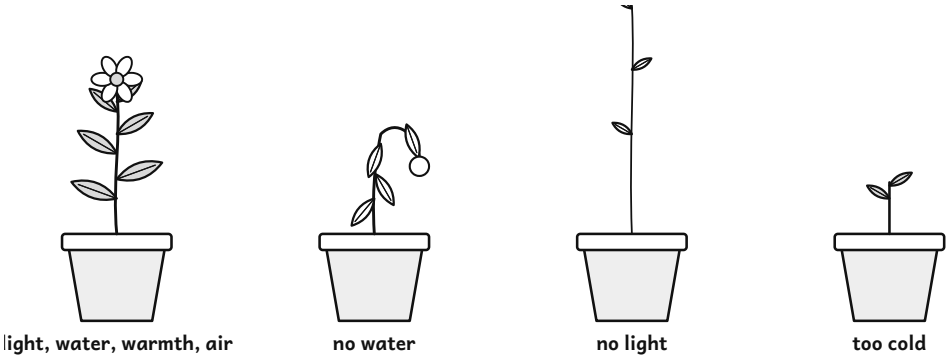
LIVING THINGS

Plants: Seeds, Germination and Growth

What is inside a seed, how seeds germinate, and what plants need to grow



In this pack

Learn it: Seeds, germination and growing plants ★☆☆
Fact check: Seeds and plant needs ★☆☆
Think it through: Explaining germination and growth ★★☆
Investigate: Does light affect plant growth? ★★☆
Real-world case study: Floating gardens: farming on flood water ★★★
Review and challenge: Show what you know ★★★

This pack belongs to: _____ **Class:** _____

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Seeds, germination and growing plants

Name: Class: Date:

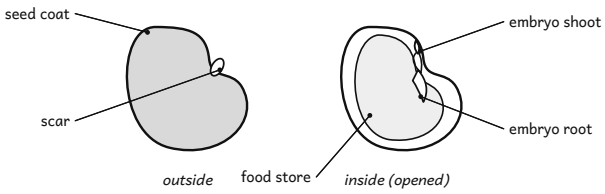
Learning goals

- Describe the parts of a seed and what each part does.
- Explain what seeds need to germinate and what plants need to grow.
- Explain why plants need light energy.

Rice, wheat, lentils, mustard, jackfruit and mango: almost every food we eat starts as a **seed**. A seed may look dry and dead, but it is alive. Inside it is a tiny plant waiting for the right moment to grow.

What is inside a seed?

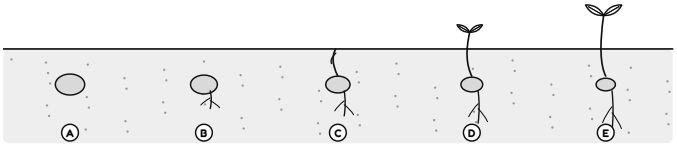
A seed has three main parts. The tough **seed coat** protects the inside from damage and from drying out. The **embryo** is the tiny baby plant, with a first root and a first shoot. The **food store** gives the embryo the energy it needs to start growing, before it has leaves to make its own food.



A bean seed, outside and opened up.

Germination

When a seed starts to grow we say it **germinates**. First the seed **absorbs** water and swells, and the seed coat splits. The first root grows **down** into the soil to take in water and hold the plant in place. Then the first shoot grows **up** towards the light. Soon the first leaves open. The seed uses up its food store and shrivels.



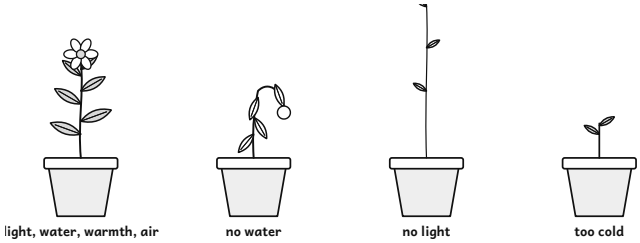
Stages A to E of a bean seed germinating.

What do seeds need to germinate?

Seeds need **water**, **warmth** and **air** (oxygen) to germinate. They do **not** need light, because the food store gives them energy. That is why seeds can germinate deep in dark soil. Seeds can wait for years until the conditions are right: lotus seeds more than 1000 years old have germinated!

What do plants need to grow?

Once a plant has leaves, it needs **light, water, warmth and air** to grow well. These are called **factors**. Most plants also take **nutrients** (minerals) from the soil through their roots.

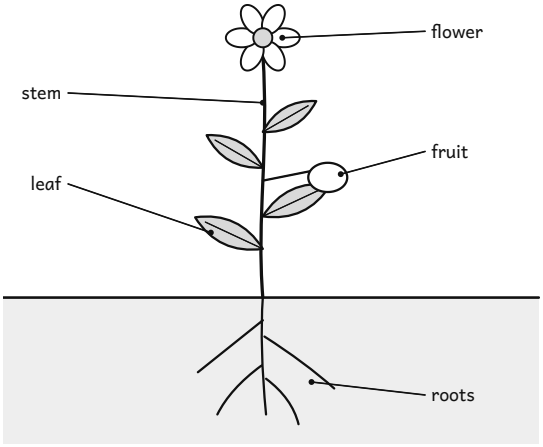


Each plant is missing one factor except the first.

Plants and light

Plants make their own food in their leaves, using **light energy** from the Sun. A green substance called **chlorophyll** traps the light. A plant kept in the dark cannot make food. It grows tall, thin and weak, its leaves turn pale yellow, and in the end it dies. Plants grow **towards** the light, which is why pot plants on a windowsill lean towards the window.

Parts of a plant and their jobs



- **Roots** take in water and nutrients and hold the plant in the ground.
- **Stem** holds the plant up and carries water and food around it.
- **Leaves** use light energy to make food.
- **Flowers** make seeds so new plants can grow.

Bangladesh connection

Bangladesh is one of the world's biggest rice producers. Farmers first grow rice seeds in a small, wet **seedbed**. After about a month the seedlings are planted out in the paddy fields.

Did you know?

A coconut is one of the biggest seeds in the world. Its "water" and white flesh are the food store for the embryo.

Key words

seed coat	tough outer layer that protects the seed
embryo	the tiny plant inside a seed
food store	part of the seed that gives the embryo energy to grow
germination	when a seed starts to grow
absorb	to take in, like a sponge takes in water
factor	something that affects how well a living thing grows
chlorophyll	green substance in leaves that traps light energy
nutrients	minerals from the soil that help plants stay healthy

Seeds and plant needs

★☆☆
Grade 5

Name: _____ Class: _____ Date: _____

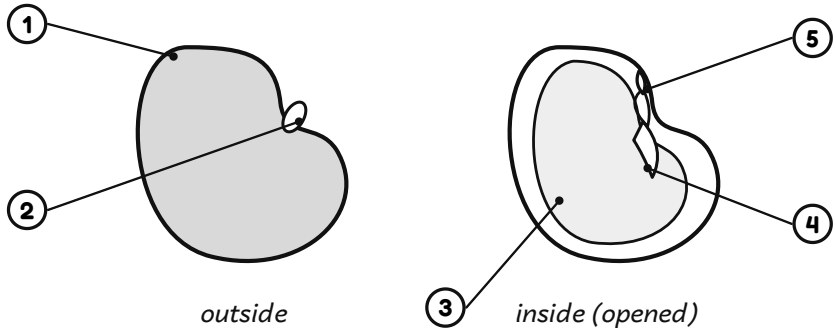
Learning goals

- Label the parts of a seed and a plant.
- Recall the conditions for germination and growth.

1 Label the parts of the bean seed. [5]

LABELS

seed coat **scar** **embryo root** **food store** **embryo shoot**



1 _____ 2 _____ 3 _____

4 _____ 5 _____

2 Complete the sentences. Use the words in the box. [7]

WORD BANK

absorbs **warmth** **flower** **germinates** **food store** **root** **soil** **shoot** **light**

When a seed _____, it first _____ water and swells. The seed coat splits and the first _____ grows downwards. Next the _____ grows upwards. The seed gets the energy it needs from its _____. Seeds need water, _____ and air to germinate, but they do not need _____.

3 Draw a line to match each plant part to its job. [4]

Draw a line to match each pair, or write the letter.

roots ●	● **A** take in water and hold the plant in the soil
stem ●	● **B** use light energy to make food
leaves ●	● **C** holds the plant up and carries water
flower ●	● **D** makes seeds for new plants

4 **True or false?** [5]

Statement	True	False	If false, correct it
1. Seeds need light to germinate.			
2. A plant kept in the dark grows pale and weak.			
3. Chlorophyll is the green substance that traps light.			
4. The embryo grows into the new plant.			
5. Plants grow away from the light.			

THINK IT THROUGH · A5 · PLANT GROWTH

Explaining germination and growth

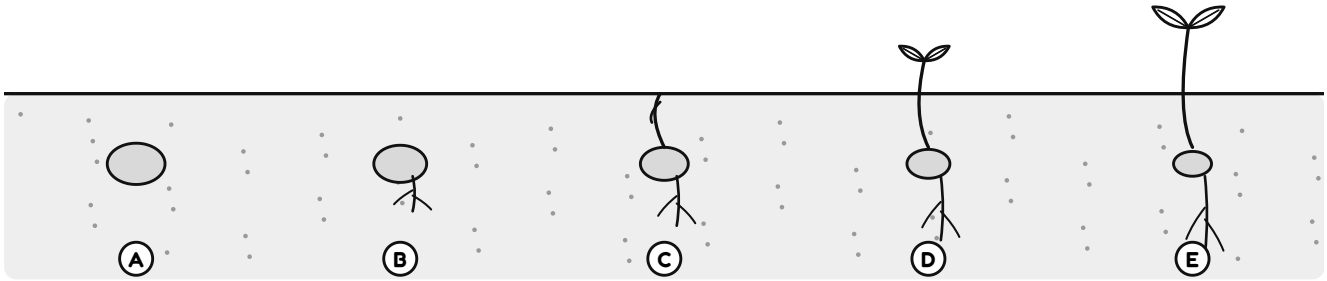
★★☆
Grade 5

Name: Class: Date:

Learning goals

- Order the stages of germination.
- Use results to decide what seeds need.
- Explain the effect of missing factors.

1 **Write the letter of each stage (A to E) next to its description.** [5]



Description	Stage
The seed has absorbed water and swollen.	
The first root grows down.	
The shoot pushes up through the soil.	
The first leaves open above the ground.	
The leaves are bigger; the seed has shrivelled.	

2 **Class 5A put 20 cress seeds in each of four dishes. After 5 days they counted how many had germinated.** [6]

Dish	Conditions	Seeds germinated (out of 20)
1	warm, wet cotton wool, in the light	18
2	warm, wet cotton wool, in a dark cupboard	19
3	cold (fridge), wet cotton wool, dark	2
4	warm, dry cotton wool, in the light	0

a) Do seeds need light to germinate? Use dishes 1 and 2 to explain.

.....

.....

b) Which dish shows that seeds need water?

.....

c) Which dish shows that seeds need warmth? Explain.

.....

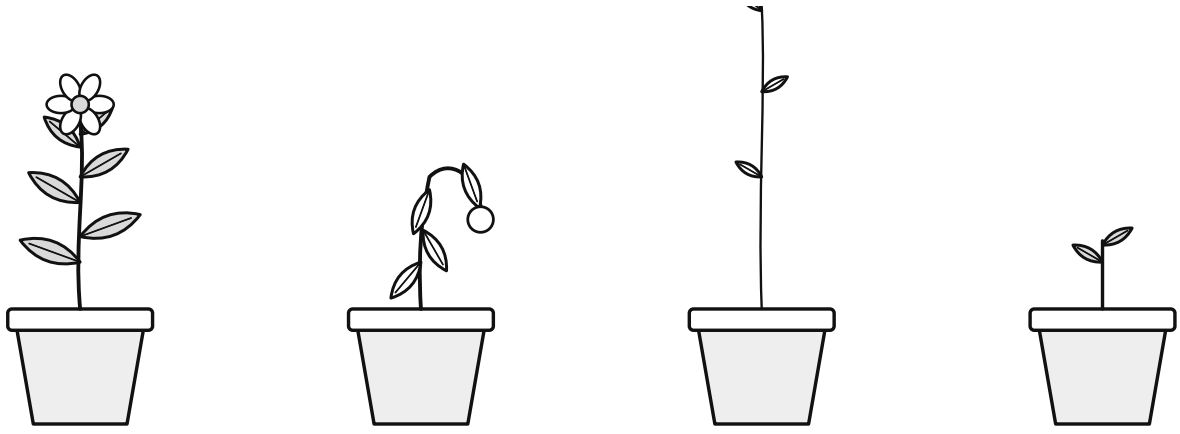
.....

d) Why did the class use 20 seeds in each dish, not just one?

.....

.....

3 **Look at the four plants. For each one, say which factor is missing and how you can tell.** [6]



light, water, warmth, air | no water | no light | too cold

Plant	Missing factor	What you can see
2		
3		
4		

4 **Why does *a* seed need *a* food store, but *a* grown plant does not?** [2]

INVESTIGATE · A5 · PLANT GROWTH

Does light affect plant growth?

★★☆ Grade 5

Name: **Class:** **Date:**

Learning goals

- Plan a fair test.
- Present results in a line graph.
- Use evidence to write a conclusion.

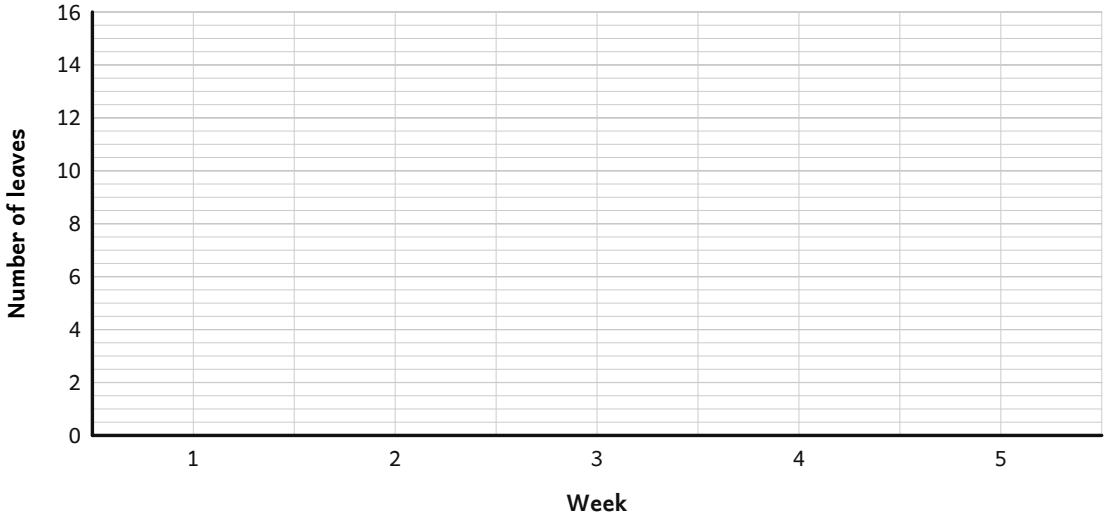
Nadia and Tanvir had two pots of the same kind of plant, each with 4 leaves. They put **Plant A** on a sunny windowsill and **Plant B** in a dark cupboard. Each week they gave both plants 50 ml of water and counted the leaves.

1 **Complete the plan for their fair test.** [5]

Question	
What we change	
What we measure	
What we keep the same	
Prediction	

2 **Here are their results. Draw a line graph with both plants on the same axes.** [5]

Week	1	2	3	4	5
Plant A: leaves (light)	4	6	9	12	15
Plant B: leaves (dark)	4	5	5	5	4



3 **Why is a line graph better than a bar chart for these results?** [1]

.....

.....

4 **What happened to Plant B in week 5? Suggest why.** [2]

.....

.....

5 **Write a conclusion for the investigation.** [2]

.....

.....

Remember

Measure from the same point each time, and repeat measurements to make your results more reliable.

Floating gardens: farming on flood water

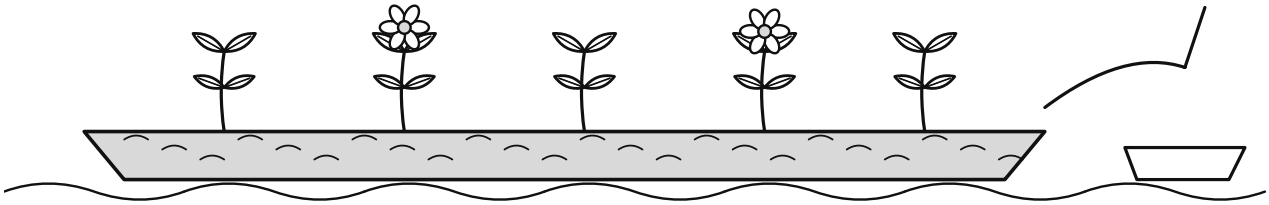
★★★ Grade 5

Name: Class: Date:

Learning goals

- Apply ideas about plant needs to a real farming method.
- Consider how people adapt to their environment.
- Link science to climate, culture and history.

Science · Geography · History · Climate change · Design



A floating garden made from water hyacinth, with vegetable seedlings growing on top.

In the south of Bangladesh, in districts such as Gopalganj, Pirojpur and Barishal, low land stays under water for several months during the monsoon. Farmers cannot plant crops in the flooded fields. For hundreds of years, families here have solved this problem with **floating gardens**, called *dhap* or *baira*.

Farmers pile up water hyacinth (a floating water plant), straw and rice stubble to make a thick raft, about one metre deep. As the plants slowly rot, they make a rich, soft compost full of **nutrients**. Farmers sow seeds of okra, gourds, red amaranth and other vegetables on top. The gardens rise and fall with the water, so the crops never drown. When the floods go down, the rotted rafts are spread on the fields as fertiliser.

In 2015 the Food and Agriculture Organization of the United Nations named Bangladesh's floating gardens a **Globally Important Agricultural Heritage System**. As climate change brings more floods, more farmers are interested in this old idea.

Sources: FAO GIAHS programme; descriptions of floating agriculture in southern Bangladesh.

1 **List the four factors that plants need to grow. Explain how a floating garden gives the plants each one.** [4]

Factor	How the floating garden provides it

2 **Why would seeds planted in a flooded field not germinate, even though there is plenty of water?** [2]

.....

.....

3 **Floating gardens use water hyacinth, a plant that grows so fast it blocks rivers. Why is this a clever idea?** [2]

.....

.....

4 **Think globally. In Mexico, the Aztecs grew food on *chinampas*, gardens built in lakes. What do people in both places have in common?** [2]

.....

.....

5 **Design. Draw a floating garden that could grow vegetables for a school. Label the materials and show how the plants get water, light, air and nutrients.** [4]

Your design

Show what you know

★★★
Grade 5

Name: Class: Date:

Learning goals

- Use key vocabulary.
- Answer test-style questions.

1 **Find the key words.**

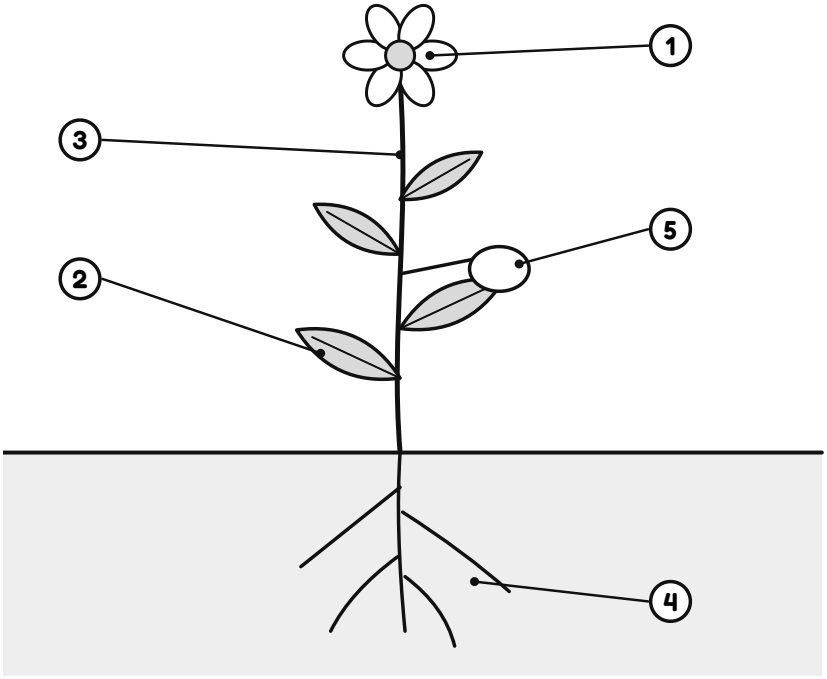
S	L	E	W	J	F	F	Y	I	A	B
F	I	F	D	U	A	S	E	E	D	E
A	B	W	B	O	C	G	M	J	U	A
L	J	A	E	P	T	L	I	G	H	T
S	I	R	E	C	O	S	J	Y	D	R
C	N	M	O	E	R	Y	L	S	E	O
S	M	T	W	T	F	W	E	T	M	O
Y	I	H	U	A	U	P	A	E	B	T
L	D	B	I	R	T	E	F	M	R	F
V	S	H	O	O	T	E	C	F	Y	H
A	G	F	P	O	T	S	R	S	O	M

embryo **factor** **leaf** **light** **root** **seed** **shoot** **stem** **warmth** **water**

2 **Label the parts of the plant.** [5]

LABELS

flower **leaf** **roots** **stem** **fruit**



1 _____ 2 _____

3 _____ 4 _____

5 _____

3 **Which part of a seed grows into the new plant?** [1]

◯ seed coat ◯ embryo

◯ food store ◯ scar

4 **A seed is planted deep in dark soil. Will it germinate? Explain.** [2]

.....

.....

5 **Challenge. Shabnam says, "Plants get their food from the soil." Is she right? Explain.** [2]

.....

.....

.....

How am I doing? Colour or tick a circle.

I can…	Not yet	Nearly	Got it!
name the parts of a seed and say what each part does	○	○	○
describe the stages of germination	○	○	○
explain that seeds need water, warmth and air, but not light, to germinate	○	○	○
explain why plants need light energy to grow	○	○	○
draw a line graph and write a conclusion	○	○	○
explain how floating gardens help people grow food in floods	○	○	○